

CompetiQuest — AI-Powered Competitive Practice System

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PROBLEM DEFINITION

Learners often rely on static question banks or out-dated quiz systems that do not adjust to their knowledge level or provide intelligent feedback. **Motivation:** To develop an AI-driven platform that offers personalized practice, feedback, and continuous improvement for students and professionals.

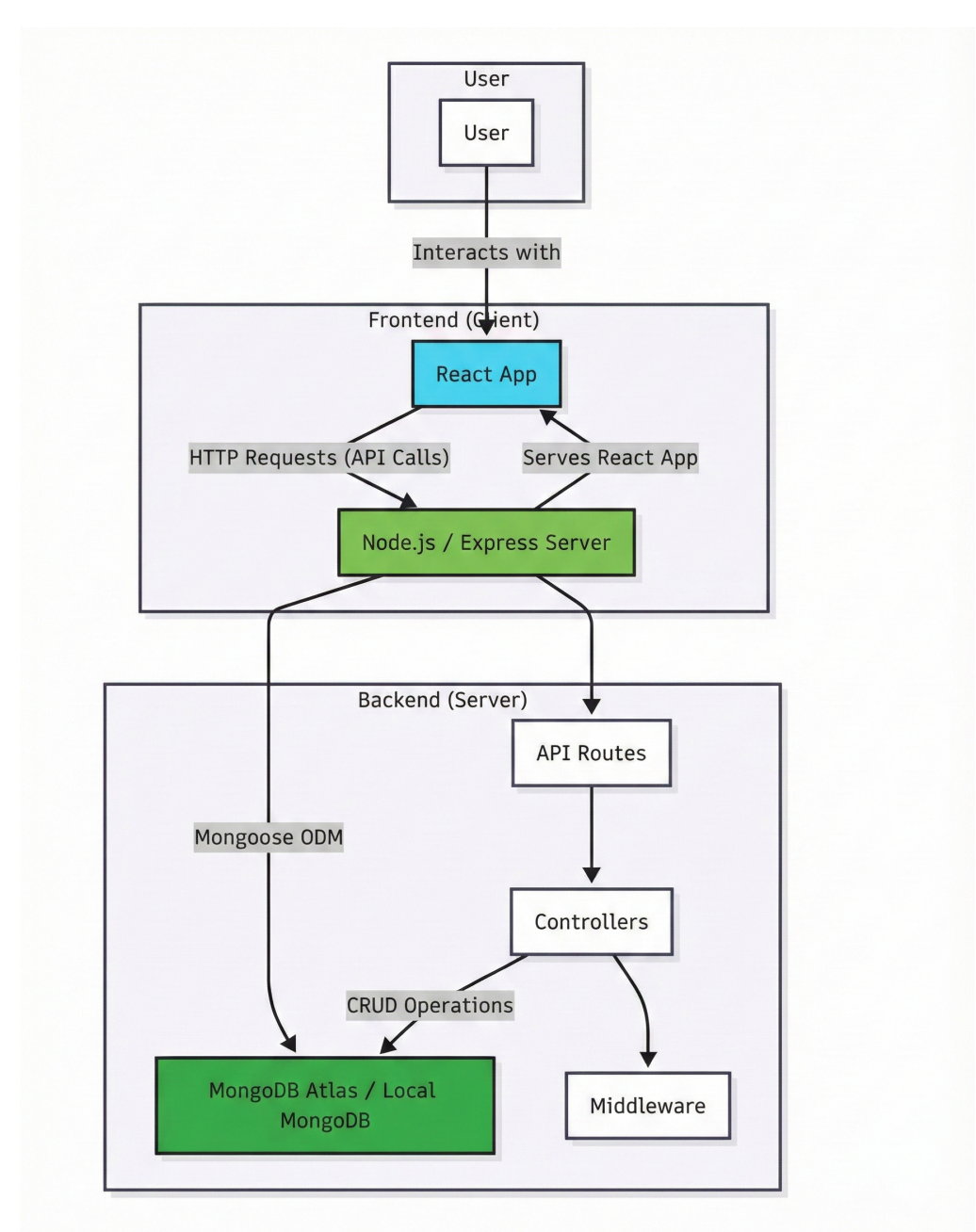
Target Users:

- Students preparing for academic or competitive exams
- Course instructors / Teachers : create guided tests and assignments

OBJECTIVES REQUIREMENTS

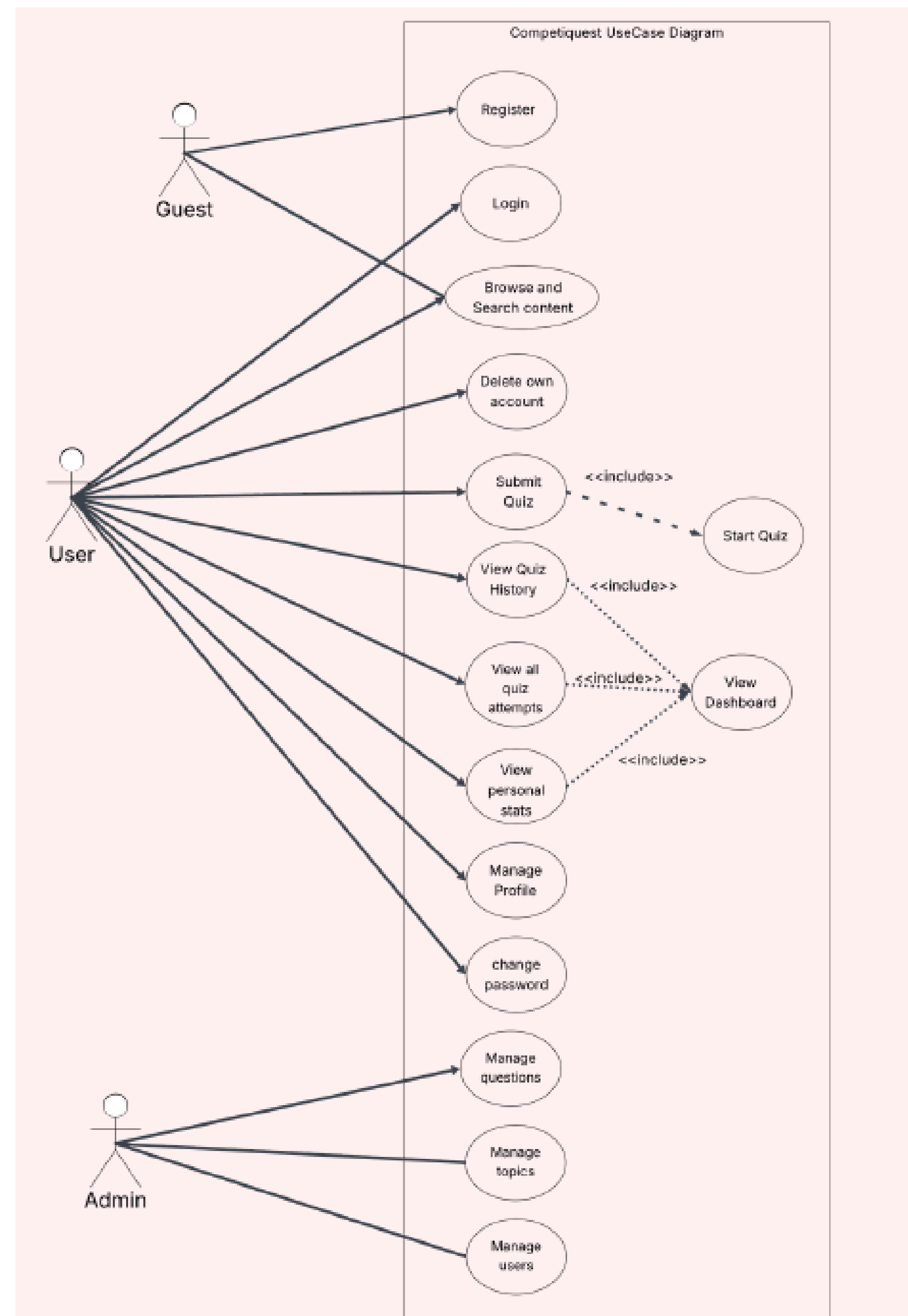
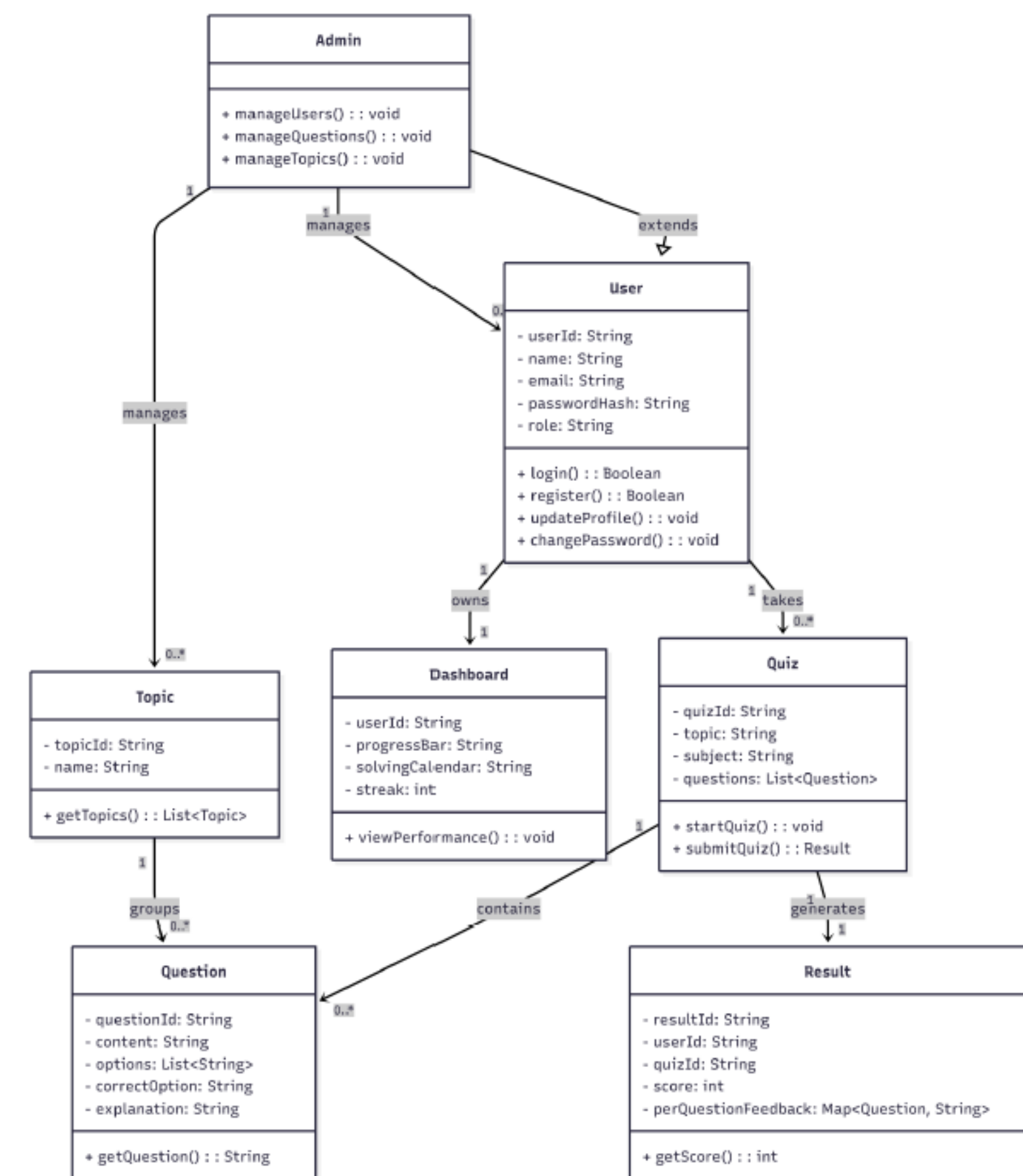
The main objective of the system is To develop an AI-powered quiz platform offering personalized learning, real-time feedback, and performance tracking. It includes role-based JWT authentication, categorized quizzes with instant scoring, a user dashboard, and an admin panel for content and user management. The system is responsive, secure, high-performing, and designed to scale and extend for future enhancements.

SYSTEM ARCHITECTURE



This architecture includes a React client that communicates with a Node.js/Express API layer to handle requests and application logic. The backend connects to MongoDB and supports middleware and scalable deployment.

DESIGN ARTIFACTS



IMPLEMENTATION HIGHLIGHTS

Tech Stack:

- Frontend:** Next.js, TailwindCSS, Framer motion, Lenis
- Backend:** Node.js, Express.js REST API
- Database:** MongoDB
- Dev Tooling:** Git/GitHub, Postman
- Deployment:** Vercel, Railway, MongoDB Atlas
- Sprint Management:** Taiga

Architectural Patterns:

- MVC, Event-driven

Key Logic:

- AI-Generated Quiz:** Fetch user performance, select questions via Gemini API, adapt difficulty, generate quiz.
- Performance Tracking:** Record answers, calculate scores, provide immediate feedback, update dashboard.
- User Management:** JWT-based authentication, role-based access, admin CRUD operations, user dashboard access.
- System Flow:** Login → Select Topic → Generate AI Quiz → Submit Answers → Evaluate → Update Dashboard

RESULTS & EVALUATION

Test Results and Performance:

Metric	Details
Total Test Suites	12 (Unit + Integration)
Total Tests Run	187
Tests Passed	185
Tests Skipped	2
Tests Failed	0
Success Rate	98.9% (100% of runnable tests)
Unit Test Time	3 seconds
Integration Test Time	9 seconds
Total Test Time	12 seconds
Consistency	All tests pass consistently.

```
Test Suites: 6 passed, 6 total
Tests: 173 passed, 173 total
Snapshots: 0 total
Time: 1.354 s
Ran all test suites matching _tests_/controllers.
```

CHALLENGES & LESSONS

- Technical Obstacles:** Integrating AI-powered quiz generation and adaptive difficulty using Gemini API.
- Testing/Debugging:** Ensuring accurate performance tracking, responsive UI, and secure JWT authentication.
- Architectural Trade-offs:** Balancing scalability with simplicity in MVC setup.
- Data Challenges:** Structuring question banks, database indexing, and optimizing query efficiency for faster response time.
- Deployment Issues:** Handling environment variables, ensuring smooth deployment.

Lessons Learned: Gained experience in AI integration, role-based access control, system scalability, and maintaining clean, modular architecture.

FUTURE WORK

- Scalability improvements for higher user load
- AI-driven feedback: identify weak topics from quizzes and suggest improvement paths
- ML integration for smarter quiz recommendations
- Web and mobile app versions for cross-platform access
- Advanced analytics for performance insights

ACKNOWLEDGEMENTS

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Links

- GitHub Repository:** <https://github.com/Aaruhsi/CompetiQuest>
- Deployment Link:** <https://competiquest-mmabxm5ck-mohsin-pathans-pr.vercel.app/>